

What is ADB?

Android Debug Bridge (ADB) is a versatile command-line tool that allows developers and security testers to communicate with an Android device (emulator or physical). It is part of the Android SDK (Software Development Kit) platform tools.

Key Features:

- Install and debug apps
- Access device shell for commands
- Transfer files between device and computer
- Access logs and system info
- Forward ports for debugging

How ADB works:

ADB runs as a client-server program:

- **Client:** runs on your development machine (your PC)
- **Server:** runs as a background process on your PC, manages connections
- **Daemon:** runs on the Android device, executes commands

Common ADB commands:

- adb devices – list connected devices

- adb shell – open command shell on the device
- adb install <apk> – install APK on device
- adb pull <remote> <local> – copy files from device
- adb push <local> <remote> – copy files to device
- adb logcat – view device logs

Security considerations:

ADB can be a security risk if enabled on a production device because it allows deep system access. Always disable it on devices not under active development or testing.

Penetration Testing Tools - Detailed Theory

Penetration Testing (PenTest) tools are software used by ethical hackers and security professionals to identify, exploit, and fix vulnerabilities in systems and applications.

Common categories:

1.

Network Scanners:

Tools that identify devices, open ports, services on a network.

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Examples: Nmap, Angry IP Scanner

2.

Vulnerability Scanners:

Automatically scan for known security weaknesses.

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Examples: Nessus, OpenVAS, Qualys

3.

Exploitation Tools:

Used to exploit vulnerabilities to gain unauthorized access.

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Examples: Metasploit Framework, Core Impact

4.

Web Application Testing Tools:

Specifically for web vulnerabilities (SQLi, XSS, CSRF).

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Examples: Burp Suite, OWASP ZAP

5.

Wireless Testing Tools:

Focus on Wi-Fi network security.

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Examples: Aircrack-ng, Kismet

6.

Password Cracking Tools:

For brute forcing passwords or recovering passwords from hashes.

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Examples: John the Ripper, Hashcat

7.

Packet Sniffers:

Capture and analyze network traffic.

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Examples: Wireshark, tcpdump

Integration with Android Testing:

Penetration testers use ADB combined with these tools to test Android apps and devices by extracting data, debugging, or simulating attacks.